

## The 6 Parent Functions

Name \_\_\_\_\_ · Aug 10

**Formula bank**    **Linear:**  $f(x) = x$     **Quadratic:**  $f(x) = x^2$     **Cubic:**  $f(x) = x^3$   
**Absolute value:**  $f(x) = |x|$     **Square root:**  $f(x) = \sqrt{x}$     **Reciprocal:**  $f(x) = \frac{1}{x}$

### VOCABULARY

**Parent function** — the \_\_\_\_\_ member of a family of functions --- no shifts, stretches, or reflections applied.

**Domain** — the set of all allowed \_\_\_\_\_ ( $x$ ) values a function can use.

**Range** — the set of all \_\_\_\_\_ ( $y$ ) values a function actually produces.

### EXAMPLE 1 · TYPE 1 — IDENTIFY FAMILY, DOMAIN, RANGE

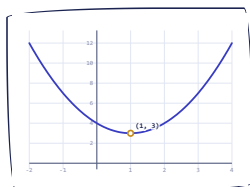
Identify the parent function of  $f(x) = \frac{1}{x}$  and give its domain and range.

1. The rule is a fraction with  $x$  in the denominator, so this is the \_\_\_\_\_ parent function.
2. A denominator can never equal 0, so  $x$  cannot be 0: domain is \_\_\_\_\_.
3. Since  $y = 1/x$  never produces an output of exactly 0 either, the range is \_\_\_\_\_.

### EXAMPLE 2 · TYPE 2 — SHIFTED EQUATION, NAME THE PARENT

Name the **ORIGINAL** parent family of  $g(x) = (x - 1)^2 + 3$ .

1. The  $-1$  and  $+3$  only \_\_\_\_\_ — they never change the family.
2. The core operation is \_\_\_\_\_, so the parent is the \_\_\_\_\_ family:  $f(x) = x^2$ .



### YOU TRY

1.

Identify the parent function of  $f(x) = x^3$  and give its domain and range.

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2.

Identify the parent function of  $f(x) = |x|$  and give its range.

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### SUMMARY

To identify a parent function: match the \_\_\_\_\_ to one of the 6 known shapes, then check for \_\_\_\_\_ caused by a root or a denominator.

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### VOCABULARY

**Parent function** — the \_\_\_\_\_ member of a family of functions --- no shifts, stretches, or reflections applied.

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### EXAMPLE 1 · TYPE 1 — IDENTIFY FAMILY, DOMAIN, RANGE

Identify the parent function of  $f(x) = \frac{1}{x}$  and give its domain and range.

1. The rule is a fraction with  $x$  in the denominator, so this is the \_\_\_\_\_ parent function.
2. A denominator can never equal 0, so  $x$  cannot be 0: domain is \_\_\_\_\_.
3. Since  $y = 1/x$  never produces an output of exactly 0 either, the range is \_\_\_\_\_.

### EXAMPLE 2 · TYPE 2 — SHIFTED EQUATION, NAME THE PARENT

Name the **ORIGINAL** parent family of  $g(x) = (x - 1)^2 + 3$ .

1. The  $-1$  and  $+3$  only \_\_\_\_\_ — they never change the family.
2. The core operation is \_\_\_\_\_, so the parent is the \_\_\_\_\_ family:  $f(x) = x^2$ .



### YOU TRY

1.

Identify the parent function of  $f(x) = x^3$  and give its domain and range.

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### SUMMARY

To identify a parent function: match the \_\_\_\_\_ to one of the 6 known shapes, then check for \_\_\_\_\_ caused by a root or a denominator.

# Meet the Parent Functions

Name: \_\_\_\_\_

Period: \_\_\_\_

Date: Aug 10

**FINAL ANSWERS — RECORD HERE**

1. \_\_\_\_\_ 2. \_\_\_\_\_ 3. \_\_\_\_\_ 4. \_\_\_\_\_ 5. \_\_\_\_\_ 6. \_\_\_\_\_ 7. \_\_\_\_\_  
8. \_\_\_\_\_ 9. \_\_\_\_\_ 10. \_\_\_\_\_ 11. \_\_\_\_\_

For each equation, name the parent function family, then state its domain and range using inequality notation.

**TYPE 1** Identify the family, domain, and range

**EXAMPLE**

**Example:**  $f(x) = \frac{1}{x}$  is the **reciprocal** parent. A denominator can't be 0, so the domain is  $x \neq 0$ ; the graph never touches  $y = 0$  either, so the range is  $y \neq 0$ .

**Directions:** Directions: Name the parent family, then give the domain and range.

1.  $f(x) = x$

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

2.  $f(x) = x^2$

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

3.  $f(x) = x^3$

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

4.  $f(x) = |x|$

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

5.  $f(x) = \sqrt{x}$

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

6.  $f(x) = \frac{1}{x}$

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

**TYPE 2** Match a shifted equation back to its parent family

**EXAMPLE**

**Example:**  $g(x) = \sqrt{x} + 4$  still has a radical, so its parent family is **square root** (the +4 just shifts it up 4).

**Directions:** Even though these are shifted, name the **ORIGINAL** parent family only.

7.  $g(x) = (x - 2)^2$

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8.  $g(x) = |x + 1| - 3$

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9.  $g(x) = \frac{1}{x - 5}$

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10.  $g(x) = (x + 3)^3 + 1$

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